

321 Architecture Virtual Circuit Board

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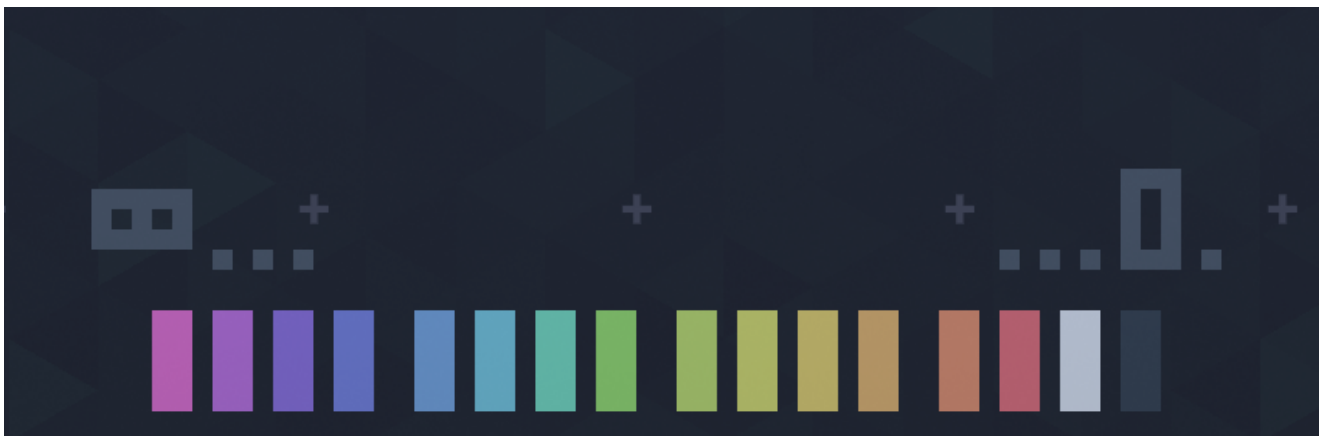
What is the 321 Architecture?

The **321 Architecture** is called this way (by me) since its based on a "von Neumann" approach with 3 different busses. Each number displays one of these busses: **3** Address Busses, **2** Control Busses, and **1** Data Bus.

Bus and Bitorder

Since the whole thing about busses in VCB is that they can transport every color at the same time, I used it to transport up to **16 bits of data** in one single bus at the same time.

For that to work good, I've used the natural color order as mine. This means **Black** is the 0 or 1 (depending on where you start counting), and then it works its way up. And since we're in Computer Science, we obviously have the **Lower Bits on the right**.



Bit Splitting Tip

Another Example of Splitting the Colors into 4-bit "parts" allowing for easier recognition of which Bit is set.



321 Architecture Explained

3 (Address Bus)

The **3** stands for the Address Busses. In total, these are 3 busses allowing 16^3 **combinations** of addresses to be allocated.

The Colors are:

- **Blue** (Largest / Most Significant, right to left)
- **Green**
- **Red** (Lower "Bits")



Address Allocation Example

This allows the Blue-Bus to function for something like a global address. For example, here **Blue-Black** is the Storage Register, while **Blue-White** are the registers used by the Fused Multiply Adder.



2 (Control Bus)

The **2** stands for the Control Bus. With 2 busses, it allows for 16^2 **combinations** of commands.

- **Light-Blue**: Larger bits
- **Purple**: Lower bits

🔗 Command Restriction

I used 2 of the light-blue commands *only* for:

1. Write into "RAM" (Puffer)
2. Read from "RAM" (Puffer)

This allows me only $14^2 + 2$ Command Combinations.



Here is the Example with the "RAM" shown:



🔗 GitHub Reference

Since I'll still change the GitHub structure, if the link does not work search for **Command Center**:

[Virtual-Circuit-Board-Projects/Virtual Circuit Board/Components/\(321Bus-Architecture\) - Command Center at main · Malte-W-Schule/Virtual-Circuit-Board-Projects](#)

1 (Data Bus)

The **1** stands for the Data Bus. It allows... well, 16^1 **combinations**.



RAM & Buffer Connection

This is directly connected to the "RAM" Buffer. Since there is only one Data Bus, mostly the inputs and outputs of components are both connected to this exact bus. Therefore, it **must have a Buffer** to store the current input to de-/activate the current component.